

Dynamic Guideline: WHO-Based Shigellosis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Current WHO status as of 29 April 2026: the latest **global WHO shigellosis-specific control guideline** identified remains *Guidelines for the control of shigellosis, including epidemics due to *Shigella dysenteriae* type 1*, published by WHO in **2005** and available through WHO publications/IRIS ([WHO publication page](#); [WHO IRIS PDF](#)). Later WHO-linked documents update or contextualize diagnosis, pediatric dysentery management, antimicrobial stewardship, outbreak risk communication, and AMR surveillance.

Important scope note on the requested “Meaningful youth engagement...” title: the WHO item found with this title is *Meaningful youth engagement in a WHO guideline development process: case study of WHO Abortion care guideline (2022)*, launched as a WHO evidence brief/webinar item on **11 December 2025**; no supplied source links this youth-engagement case study to shigellosis or *Shigella* guidance ([WHO webinar page](#); [WHO PDF](#)). Therefore, it is **not treated as a shigellosis clinical guideline**.

Exactly 5 WHO guideline/guidance documents synthesized for this report

No.	WHO document	Publication / update status from retrieved sources	Target population	Scope and relevance to shigellosis
1	<i>Guidelines for the control of shigellosis, including epidemics due to <i>Shigella dysenteriae</i> type 1</i>	2005; exact day/month not extractable from retrieved source	All patients; outbreak settings; special focus on <i>S. dysenteriae</i> type 1	Core WHO shigellosis control guideline; WHO page states fluoroquinolones as first-line treatment and zinc should also be given (WHO publication page).
2	<i>Pocket book of hospital care for children: second edition</i>	Second edition; WHO describes it as update of 2005 edition; cited as 2013 by MSF	Children at first-level referral hospitals	IMCI-linked hospital guidance for common childhood illnesses, including dysentery/shigellosis management context (WHO Pocket Book ; MSF guideline citing WHO Pocket Book).

No.	WHO document	Publication / update status from retrieved sources	Target population	Scope and relevance to shigellosis
3	<i>Recommendations for management of common childhood conditions: evidence for technical update of Pocket Book recommendations</i>	2012; exact day/month not extractable	Children, especially in low- and middle-income countries	WHO evidence update for dysentery treatment; evaluated WHO-recommended antibiotics: ciprofloxacin, ceftriaxone, pivmecillinam, and azithromycin (WHO 2012 evidence update via NCBI).
4	WHO AWaRe — Access, Watch, Reserve — antibiotic book	WHO publication; AWaRe classification created 2017 and revised 2019/2021; book page lists newer WHO infectious-disease guidelines	Children and adults	WHO antibiotic stewardship framework. The AWaRe book is not itself a formal WHO guideline , but its infection-specific antibiotic choices represent formal recommendations (WHO AWaRe antibiotic book ; WHO Bulletin article).
5	<i>Guideline on management of pneumonia and diarrhoea in children up to 10 years of age</i>	Listed by WHO AWaRe page as published December 2024; exact day not extractable	Children up to 10 years	Most recent WHO child diarrhoea guideline identified in retrieved sources; detailed Shigella-specific recommendation text was not available in retrieved snippets (WHO AWaRe antibiotic book).

1.2 Key WHO-Based Recommendations

The recommendations below synthesize the 2005 WHO shigellosis guideline with subsequent WHO-linked guidance and post-2005 evidence. Because the retrieved sources did not verify that the 2005 WHO guideline itself used GRADE, the **GRADE ratings below are this report’s evidence-to-decision judgments**, using standard GRADE concepts: risk of bias, consistency, directness, precision, publication bias, benefits/harms, feasibility, equity, acceptability, and AMR impact.

Recommendation	GRADE Evidence Quality	Strength
Recognize suspected shigellosis/dysentery clinically as diarrhoea with visible blood in stool , particularly in outbreaks or low-resource settings where	Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
laboratory testing is delayed or unavailable.		
Confirm diagnosis with stool culture when feasible , because culture remains the diagnostic gold standard and enables serotyping and susceptibility testing.	Moderate	Strong
Use culture-independent diagnostic tests only as adjuncts where available; confirm positive CIDT results with culture when treatment or public-health action depends on isolate characterization.	Moderate	Strong
Provide oral rehydration, hydration monitoring, and electrolyte support for all patients; use IV fluids/hospitalization for severe dehydration, severe illness, or inability to drink.	High	Strong
For mild shigellosis, do not routinely use antibiotics , because most patients recover within 5–7 days and unnecessary antibiotics increase AMR selection pressure.	Moderate	Strong
Consider antibiotics promptly for acute bloody diarrhoea/dysentery, severe disease, hospitalized patients, suspected <i>S. dysenteriae</i> type 1, immunocompromised patients, bacteremia/sepsis, high-risk transmission occupations, and outbreak-control situations.	Moderate	Strong
When antibiotics are used, select therapy using local susceptibility data or the patient/outbreak isolate’s antimicrobial susceptibility testing whenever possible.	High	Strong
Do not rely automatically on the 2005 WHO fluoroquinolone-first approach in settings with substantial ciprofloxacin resistance; update empiric therapy to local AMR epidemiology.	High	Strong
Potential antibiotic options, depending on age, pregnancy status, severity, route feasibility, local guidance, and AST, include ciprofloxacin, ceftriaxone, azithromycin, and pivmecillinam where available.	Moderate	Conditional
Avoid antimotility drugs such as loperamide and diphenoxylate-atropine in suspected or confirmed shigellosis/dysentery.	Low	Strong
Implement WASH-based prevention: hand hygiene, safe water, sanitation, safe food handling, disinfection, isolation/exclusion policies where appropriate, and community risk communication.	Moderate	Strong
Report suspected outbreaks and notifiable cases according to national requirements; use early detection, investigation, situation reporting, and coordinated outbreak response.	Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
Strengthen AMR surveillance and antimicrobial stewardship because drug-resistant <i>Shigella</i> is a WHO high-priority AMR pathogen and GLASS 2025 reports substantial ciprofloxacin resistance.	High	Strong
Do not recommend routine <i>Shigella</i> vaccination outside clinical trials or policy-approved programs as of 29 April 2026 , because no licensed vaccine or definitive phase 3 efficacy evidence was identified in the retrieved evidence.	Low	Conditional

1.3 Guideline Development Methodology

Evidence review process

- The 2005 WHO shigellosis guideline is the central disease-specific global WHO document located ([WHO publication page](#)).
- Later WHO evidence work for childhood dysentery included systematic review of antibiotics currently recommended by WHO—ciprofloxacin, ceftriaxone, pivmecillinam, and azithromycin—in children under 16 years in low- and middle-income countries ([WHO 2012 evidence update via NCBI](#)).
- WHO-linked surveillance and outbreak-control guidance includes:
 - WHO AFRO Integrated Disease Surveillance and Response case definitions for dysentery and confirmed *S. dysenteriae* type 1 ([WHO AFRO IDSR technical guidelines](#)).
 - WHO EMRO *Shigella* risk communication and community engagement guidance emphasizing clinical recognition in resource-limited settings, hydration, selective antibiotics, hygiene, sanitation, and AMR stewardship ([WHO EMRO guidance](#)).
 - WHO GLASS 2025 global AMR surveillance, including *Shigella* spp. gastrointestinal infections ([WHO GLASS 2025 publication page](#); [WHO GLASS 2025 PDF](#)).

Consensus method

- The retrieved sources do not provide a directly citable description of the 2005 WHO shigellosis guideline’s full panel process, voting procedure, or consensus method.
- The WHO AWaRe antibiotic book is described as closely aligned with WHO Essential Medicines Lists and AWaRe antibiotic classification; its infection-specific antibiotic choices are formal recommendations, although the book itself is not a formal WHO guideline ([WHO Bulletin article](#)).

Conflict of interest management

- The retrieved summaries do not provide document-specific conflict-of-interest details for the 2005 WHO shigellosis guideline.
- For this report, evidence certainty was downgraded where guideline-development details, direct GRADE methods, or conflicts-of-interest processes were not verifiable from retrieved sources.

II. Evidence Summary After WHO Latest Guideline Release (Guidelines for the control of shigellosis, including epidemics due to *Shigella dysenteriae* type 1) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was prioritized in the following order:

1. WHO global guidelines, WHO regional technical documents, WHO IRIS/publication pages, WHO GLASS, and WHO AWaRe resources.
2. National and regional public-health authorities: CDC, NIH HIV opportunistic infection guidance, and public-health operational guidance.
3. Systematic reviews and meta-analyses of dysentery/shigellosis treatment.
4. Randomized controlled trials of antibiotic therapy.
5. AMR surveillance reports and outbreak alerts.
6. Vaccine trial and trial-design literature.

Databases and sources searched

- WHO publications, WHO IRIS, WHO EMRO, WHO AFRO, WHO GLASS, WHO AWaRe.
- PubMed/NCBI Bookshelf/PMC.
- CDC clinical guidance, CDC antimicrobial-resistance pages, CDC MMWR.
- NIH HIV opportunistic infection guideline via NCBI.
- Peer-reviewed journal sources and trial reports indexed in PubMed/PMC.
- Clinical and public-health reviews where primary guidance was not available in full.

Inclusion criteria

- Human shigellosis, dysentery, or *Shigella* infection.
- Diagnosis, treatment, antimicrobial resistance, prevention, outbreak management, or vaccine evidence.
- WHO-linked guidance and authoritative post-2005 evidence through **29 April 2026**.
- Systematic reviews, randomized trials, surveillance reports, and major public-health guidance.

Exclusion criteria

- Non-human-only studies.
- Narrative claims without retrievable source support.
- Vaccine or antimicrobial claims without extractable efficacy/safety/resistance information.
- The WHO youth-engagement evidence brief was excluded from clinical recommendation development because it concerns the WHO Abortion care guideline, not shigellosis ([WHO webinar page](#)).

2.2 Key New Studies and Guidance After 2005

Study / source	Design	Population / setting	Key Findings	GRADE Quality
WHO 2012 Pocket Book evidence update for dysentery	Evidence review / technical update	Children <16 years, mainly LMICs	Evaluated WHO-recommended antibiotics: ciprofloxacin, ceftriaxone, pivmecillinam, and azithromycin; short 3–5 day therapy considered beneficial for dysentery (WHO 2012 evidence update).	Moderate
Systematic review of dysentery/shigellosis treatment guidelines	Systematic review of reviews/RCTs	Human shigellosis/dysentery evidence since 2005	Reported WHO 2005 ciprofloxacin pediatric first-line dose 15 mg/kg orally twice daily for 3 days; noted pivmecillinam and ceftriaxone as second-line options and rising AMR as a major challenge (PMC review).	Moderate
CHERG pediatric review summarized in guideline review	Systematic review of RCTs	Children <16 years, mainly LMICs	Ciprofloxacin, pivmecillinam, or ceftriaxone reduced clinical failure by 82%; Shigella pathogen clearance 96% (PMC review).	Moderate
2010 systematic review summarized in guideline review	Systematic review	Children in LMIC hospital settings	Ciprofloxacin, pivmecillinam, and ceftriaxone effective clinically and bacteriologically; reported clinical failure rate 0.1% (PMC review).	Moderate
Gatifloxacin vs ciprofloxacin RCT, Vietnam	Multicenter randomized open-label controlled trial	494 children with uncomplicated dysentery/shigellosis; 107 culture-positive for Shigella	Gatifloxacin was not superior to ciprofloxacin; clinical failure 12.0% vs 11.0%, p=0.72; similar	Moderate

Study / source	Design	Population / setting	Key Findings	GRADE Quality
			symptom duration (PubMed ; PLOS NTD).	
Azithromycin vs ciprofloxacin RCT, Bangladesh	Double-blind RCT	70 adult men with shigellosis ≤72 hours	Clinical success 82% with azithromycin vs 89% with ciprofloxacin; bacteriologic success 94% vs 100%; azithromycin effective against multidrug-resistant <i>Shigella</i> in this setting (PubMed).	Low
CDC Clinical Overview	Public-health clinical guidance	United States; general clinical practice	Recommends stool testing by culture or CIDT, culture confirmation after CIDT, AST when antibiotics are planned, and susceptibility-guided treatment; lists ciprofloxacin, azithromycin, and ceftriaxone as recommended options when treatment is needed (CDC Clinical Overview).	Moderate
CDC AMR <i>Shigella</i> guidance	Public-health guidance	United States	Most people recover without antibiotics; antimicrobial-resistant <i>Shigella</i> infections rising since 2016; estimated 242,000 resistant infections annually in the U.S. (CDC AMR Shigella).	Moderate
CDC MMWR 2026: XDR shigellosis	National surveillance report	U.S. isolates, 2011–2023	XDR defined as resistance to ampicillin, azithromycin, ceftriaxone, ciprofloxacin, and TMP-SMX; XDR proportion increased; no optimal therapy established; AST-	Moderate

Study / source	Design	Population / setting	Key Findings	GRADE Quality
			guided therapy and timely reporting emphasized (CDC MMWR).	
CDC/IDSA alert on ciprofloxacin and azithromycin failures	Public-health alert	Patients treated with fluoroquinolones or azithromycin	Increasing isolates with ciprofloxacin resistance mechanisms despite susceptible MIC range; elevated azithromycin MICs; monitor, obtain AST if failure suspected, consult ID, report failures (IDSA-hosted CDC alert).	Low
NIH HIV opportunistic infection guidance	Specialty clinical guideline	People with HIV	Treating Shigella may slightly shorten illness and reduce transmission; select antibiotics by susceptibility; in mild resolving diarrhea with CD4 >500, antibiotics may be withheld; severe illness may require empiric parenteral carbapenem pending susceptibilities (NIH/NCBI HIV OI guideline PDF).	Low
WHO EMRO Shigella RCCE guidance	WHO regional operational guidance	Outbreaks, resource-limited settings, vulnerable populations	Mild cases often need fluids/rest; most recover without antibiotics in 5–7 days; consider initial antibiotics for acute bloody diarrhoea and immunocompromised patients; avoid loperamide/diphenoxylate; emphasize WASH and AMR stewardship (WHO EMRO guidance).	Low
WHO AFRO IDSR technical guidelines	Surveillance/outbreak guidance	African Region; surveillance systems	Suspected dysentery: diarrhoea with visible	Moderate

Study / source	Design	Population / setting	Key Findings	GRADE Quality
			blood; confirmed: stool culture positive for <i>S. dysenteriae</i> type 1; includes outbreak response, situation reporting, case treatment, and disinfection (WHO AFRO IDSR).	
WHO Shigella vaccine preferred product characteristics draft	WHO vaccine-development technical document	Vaccine development	States conventional bacterial culture is the gold standard for Shigella diagnosis and enables serotyping for vaccine trial endpoints (WHO draft PPC PDF).	Low
WHO GLASS 2025	Global AMR surveillance	104 countries reporting 2023 data; 110 countries 2016–2023	Includes Shigella spp.; global ciprofloxacin resistance in Shigella gastrointestinal infections in 2023 estimated at 29.7% and 75.5% in WHO South-East Asia Region (WHO GLASS 2025).	Moderate
WHO BPPL 2024	WHO AMR priority-setting document	Global AMR R&D prioritization	Shigella listed among high-priority drug-resistant bacterial pathogens; high burden in LMICs noted (WHO BPPL 2024).	Moderate
1790GAHB Shigella sonnei GMMA vaccine trial	Phase 2b randomized placebo-controlled human challenge study	Healthy adults	Evaluated efficacy, safety, and immunogenicity after two doses and oral challenge; snippets did not provide numerical efficacy or safety estimates (PubMed).	Low
Shigella vaccine pivotal trial-design working group	Expert trial-design guidance	Young children in LMICs	Proposed pivotal endpoint: first episode of acute moderate/severe	Very Low

Study / source	Design	Population / setting	Key Findings	GRADE Quality
			diarrhea due to vaccine-included Shigella strains (PubMed).	
Shigella vaccine review	Narrative/review	Vaccine-development landscape	Describes conjugate, recombinant/chimeric, live attenuated, and newer genomics-informed approaches; no licensed vaccine identified from retrieved evidence (MDPI review).	Low

2.3 Clinical Synthesis by Domain

A. Diagnosis

WHO-aligned diagnostic approach

- Suspect shigellosis/dysentery** in a patient with acute diarrhoea plus visible blood, abdominal cramps, fever, tenesmus, or outbreak exposure.
- Assess dehydration and severity immediately.**
- Obtain stool testing** when feasible:
 - Stool culture is the gold standard and enables serotyping and AST ([WHO draft PPC PDF](#)).
 - CDC supports stool culture or CIDT, with culture confirmation after positive CIDT when isolate characterization is needed ([CDC Clinical Overview](#)).
- Perform AST whenever antibiotics are planned**, especially in severe disease, outbreaks, immunocompromised patients, treatment failure, recent travel, men who have sex with men, homelessness, childcare exposure, or suspected XDR Shigella.

GRADE diagnostic table

Diagnostic action	Rationale	GRADE Quality	Strength
Use visible blood in stool as a syndromic trigger for dysentery/shigellosis surveillance and treatment decisions.	WHO AFRO IDSR suspected case definition; practical in low-resource settings.	Moderate	Strong
Prefer stool culture where feasible.	Culture confirms Shigella and enables serotyping/AST.	Moderate	Strong
Confirm positive CIDT with culture when treatment or public-health action depends on	Needed for AST and outbreak characterization.	Moderate	Strong

Diagnostic action	Rationale	GRADE Quality	Strength
isolate data.			
Obtain AST before or during antibiotic therapy whenever possible.	Rising AMR and XDR Shigella make empiric therapy unreliable.	High	Strong

B. Supportive care

Supportive care remains central. WHO EMRO notes most individuals recover without antibiotics within 5–7 days and mild disease often requires fluids and rest ([WHO EMRO guidance](#)). StatPearls similarly describes hydration and electrolyte management as the mainstay, with IV fluids/hospitalization for severe cases ([NCBI Bookshelf/StatPearls](#)).

Supportive-care recommendation	GRADE Quality	Strength
Give oral rehydration solution or appropriate fluids to all patients at risk of dehydration.	High	Strong
Use IV fluids and hospital care for severe dehydration, shock, inability to drink, severe systemic illness, or high-risk infants/older adults/immunocompromised patients.	High	Strong
Continue feeding as tolerated; monitor children closely for dehydration and malnutrition.	Moderate	Strong
Give zinc to children where consistent with WHO diarrhoea guidance and local policy; WHO 2005 shigellosis page states patients should receive zinc.	Moderate	Strong
Avoid antimotility agents such as loperamide and diphenoxylate-atropine.	Low	Strong

C. Antibiotic treatment

Key change since WHO 2005: WHO’s 2005 page states fluoroquinolones should be used as first-line treatment for all shigellosis patients ([WHO publication page](#)). However, WHO GLASS 2025 reports substantial global ciprofloxacin resistance in *Shigella* spp. gastrointestinal infections—**29.7% globally** and **75.5% in the WHO South-East Asia Region** in 2023 ([WHO GLASS 2025 PDF](#)). Therefore, the 2005 fluoroquinolone-first recommendation should now be interpreted through local AMR data and AST.

When to treat with antibiotics

Antibiotics should be considered or given when any of the following are present:

- Acute bloody diarrhoea/dysentery, especially moderate–severe disease.
- Suspected or confirmed *S. dysenteriae* type 1.
- Severe illness, hospitalization, sepsis, bacteremia, or high fever/toxicity.
- Immunocompromised state, including advanced HIV or chemotherapy.
- Infants, frail older adults, or high-risk clinical context.

- Food handlers, healthcare workers, childcare workers, institutional settings, or outbreak-control need.
- Persistent symptoms or high-risk transmission settings.

Antibiotic options

Antibiotic	Role	Evidence and cautions	GRADE Quality	Strength
Ciprofloxacin	Historical WHO first-line; still possible if local susceptibility supports use	WHO 2005 and later reviews cite ciprofloxacin first-line; AMR now substantial; CDC warns of possible treatment failures despite “susceptible” MICs when resistance mechanisms exist (PMC review ; IDSA-hosted CDC alert).	Moderate	Conditional
Ceftriaxone	Severe disease, inability to take oral therapy, children, pregnancy depending on local guidance and AST	WHO-linked reviews list ceftriaxone as alternative/second-line; CDC lists third-generation cephalosporins such as ceftriaxone when treatment is needed (CDC Clinical Overview).	Moderate	Conditional
Azithromycin	Alternative where susceptible; often oral option	Adult RCT showed similar clinical success to ciprofloxacin but small/old; CDC warns elevated azithromycin MICs and possible failure (PubMed ; IDSA-hosted CDC alert).	Low	Conditional
Pivmecillinam / amdinocillin pivoxil	Alternative where available and susceptible	WHO 2005-linked reviews list as second-line; availability varies.	Moderate	Conditional
Carbapenem	Severe suspected XDR disease, especially immunocompromised, pending AST and specialist input	NIH HIV guidance recommends empiric parenteral carbapenem for severely ill people with HIV pending susceptibilities (NIH/NCBI HIV OI guideline PDF).	Low	Conditional

Antibiotic	Role	Evidence and cautions	GRADE Quality	Strength
Chloramphenicol, fosfomycin, oral carbapenems such as sulopenem	Not routine standard therapy	CDC MMWR states chloramphenicol is not routinely recommended in the U.S.; pivmecillinam, fosfomycin, and oral carbapenems might be effective but were not FDA-approved for shigellosis as of 2025 (CDC MMWR).	Very Low	Conditional / specialist-only

D. AMR and XDR Shigella

AMR is now central to shigellosis management.

- WHO’s 2024 Bacterial Priority Pathogens List identifies Shigella among **high-priority** drug-resistant pathogens, with high burden in LMICs ([WHO BPPL 2024](#)).
- WHO GLASS 2025 includes Shigella spp. and reports 2023 ciprofloxacin resistance of **29.7% globally** and **75.5% in South-East Asia** ([WHO GLASS 2025 PDF](#)).
- CDC’s 2026 MMWR defines XDR Shigella as resistant to **ampicillin, azithromycin, ceftriaxone, ciprofloxacin, and TMP-SMX**, or having WGS resistance determinants for these agents; CDC emphasizes AST-guided therapy and timely reporting ([CDC MMWR](#)).

AMR recommendation	GRADE Quality	Strength
Use AST-guided therapy whenever possible.	High	Strong
Maintain local/regional Shigella susceptibility surveillance and share data rapidly during outbreaks.	High	Strong
Avoid unnecessary antibiotics in mild self-limited illness.	Moderate	Strong
Monitor for treatment failure after ciprofloxacin or azithromycin and repeat stool culture/AST if failure occurs.	Moderate	Strong
Consult infectious-disease/public-health specialists for suspected XDR Shigella.	Low	Strong

E. Outbreak control and prevention

WHO AFRO IDSR guidance supports early detection, investigation, response, situation reporting, case treatment during outbreaks, and disinfection ([WHO AFRO IDSR](#)). WHO emergency medical team standards emphasize coordinated roles, real-time information sharing, strategic outbreak preparedness and response plans, and pillars such as case management, logistics, and risk communication/community engagement ([WHO EMT minimum](#)

[standards](#)). WHO EMRO RCCE guidance emphasizes symptoms, transmission, hygiene, safe water/food, sanitation, and community education across all age groups, especially vulnerable populations ([WHO EMRO guidance](#)).

Outbreak-control recommendation	GRADE Quality	Strength
Activate surveillance for bloody diarrhoea clusters and report according to national rules.	Moderate	Strong
Investigate common food/water sources and person-to-person transmission chains.	Moderate	Strong
Implement handwashing with soap, safe water, sanitation, safe food handling, and environmental cleaning/disinfection.	Moderate	Strong
Provide targeted risk communication for childcare settings, schools, displaced populations, institutions, food handlers, MSM networks where relevant, and immunocompromised populations.	Low	Strong
Coordinate logistics, case management, laboratory testing, AMR surveillance, and community engagement as formal outbreak pillars.	Low	Strong

F. Vaccines

No licensed *Shigella* vaccine or definitive phase 3 efficacy evidence was identified in the retrieved sources. Vaccine development is active, including GMMA, conjugate, live attenuated, recombinant, and genomics-informed platforms ([MDPI review](#)). A phase 2b human challenge study evaluated the *S. sonnei* GMMA vaccine candidate 1790GAHB, but retrieved snippets did not provide numerical efficacy/safety results ([PubMed](#)). Trial-design experts propose pivotal endpoints based on prevention of first acute moderate/severe diarrhea caused by vaccine-included *Shigella* strains in young children in LMICs ([PubMed](#)).

Vaccine recommendation	GRADE Quality	Strength
Do not use <i>Shigella</i> vaccines routinely outside clinical trials or approved public-health programs as of 29 April 2026.	Low	Conditional
Prioritize participation in well-designed trials in high-burden settings, especially pediatric LMIC populations.	Low	Conditional
Use culture/molecular confirmation and clinically meaningful diarrhea endpoints in vaccine trials.	Low	Conditional

2.3 Evidence Gaps and Future Research Needs

- Updated WHO global shigellosis guideline needed.** The core WHO shigellosis-specific guideline is from 2005, while AMR patterns have changed substantially.
- Empiric antibiotic regimens need regional updating.** Ciprofloxacin resistance is high in some regions, especially South-East Asia, making old first-line recommendations unsafe if applied without AST/local data.

3. **XDR treatment evidence is weak.** CDC notes no optimal therapy has been established for XDR shigellosis ([CDC MMWR](#)).
4. **More pediatric RCTs are needed** comparing azithromycin, ceftriaxone, pivmecillinam, ciprofloxacin, and newer oral agents in regions with high AMR.
5. **Better AMR surveillance coverage is needed**, particularly in LMICs and outbreak settings.
6. **Rapid diagnostic and resistance tests** need evaluation for clinical outcomes, outbreak control, and stewardship impact.
7. **Vaccine efficacy remains unproven at policy level.** Phase 3 pediatric efficacy trials in high-burden settings are needed.
8. **Long-term outcomes** such as stunting, growth faltering, and enteric dysfunction after Shigella prevention remain uncertain; expert reports identify these as critical but difficult endpoints ([PMC expert report](#); [medRxiv preprint](#)).
9. **Guideline-development inclusivity:** WHO's youth-engagement evidence brief concerns abortion-care guidance, not shigellosis; future shigellosis guidelines could still incorporate youth/community engagement principles for outbreak communication, school-based prevention, and adolescent/young adult transmission settings ([WHO webinar page](#)).

2.4 Updated Recommendations Based on New Evidence

Original WHO 2005 recommendation / position	New Evidence Impact	Updated Recommendation as of 29 April 2026
Fluoroquinolones should be first-line treatment for all shigellosis patients (WHO publication page).	WHO GLASS 2025 reports substantial ciprofloxacin resistance; CDC reports rising AMR/XDR and treatment-failure concerns.	Use ciprofloxacin only when local susceptibility or AST supports it. Do not use a universal fluoroquinolone-first approach in high-resistance settings.
Treat shigellosis with antibiotics.	WHO EMRO and CDC note most mild cases recover without antibiotics in 5–7 days; AMR pressure is substantial.	Reserve antibiotics for dysentery, severe disease, high-risk patients, immunocompromised patients, outbreak-control needs, or transmission-risk occupations.
Ciprofloxacin 15 mg/kg orally twice daily for 3 days as first-line pediatric treatment, as reported in later review of WHO 2005 guidance (PMC review).	Pediatric systematic reviews support efficacy of ciprofloxacin/ceftriaxone/pivmecillinam, but AMR reduces reliability.	For children, select ciprofloxacin, ceftriaxone, azithromycin, or pivmecillinam according to severity, route feasibility, contraindications, and AST/local AMR.
Pivmecillinam and ceftriaxone as second-line options when	Evidence supports efficacy; availability and resistance vary.	Use ceftriaxone for severe disease or inability to take oral therapy; use pivmecillinam

Original WHO 2005 recommendation / position	New Evidence Impact	Updated Recommendation as of 29 April 2026
ciprofloxacin unsuitable or resistance known.		where available and susceptible.
Zinc should be given.	Remains consistent with pediatric diarrhoea care and WHO 2005 page.	Give zinc to children with diarrhoea/dysentery where consistent with WHO/local pediatric diarrhoea policy.
Outbreak control focused on case treatment and control of <i>S. dysenteriae</i> type 1 epidemics.	WHO AFRO IDSR, WHO EMRO RCCE, and WHO EMT standards emphasize integrated surveillance, WASH, coordination, logistics, and communication.	Use a full outbreak package: early detection, laboratory confirmation/AST, case management, WASH, disinfection, safe food/water, risk communication, situation reporting, and coordinated logistics.
No vaccine recommendation.	Vaccine candidates are progressing, but no licensed vaccine or definitive phase 3 efficacy result identified.	No routine <i>Shigella</i> vaccine recommendation outside trials or approved programs.

Practical WHO-Based Clinical Algorithm

1. Patient presents with diarrhoea

- Assess dehydration, sepsis, age, pregnancy, immune status, outbreak exposure.

2. If visible blood in stool or severe symptoms

- Treat as suspected dysentery/shigellosis.
- Obtain stool culture and AST if feasible.
- Notify public health if outbreak, high-risk setting, or notifiable disease requirement.

3. Supportive care for all

- ORS/fluids; IV fluids if severe.
- Continue nutrition.
- Zinc for children per local/WHO diarrhoea policy.
- Avoid loperamide/diphenoxylate-atropine.

4. Antibiotics

- Not routine for mild watery/self-limited disease.
- Use for dysentery, severe disease, immunocompromise, sepsis/bacteremia, suspected *S. dysenteriae* type 1, high-risk occupations, or outbreak control.
- Choose agent using AST/local AMR:
 - Ciprofloxacin if susceptible and appropriate.
 - Ceftriaxone for severe disease, inability to take oral therapy, or where indicated.

- Azithromycin if susceptible/appropriate.
- Pivmecillinam where available and susceptible.
- Specialist input for suspected XDR disease.

5. If no improvement within 48–72 hours or relapse

- Reassess diagnosis.
- Repeat stool culture/AST.
- Evaluate adherence and vomiting/absorption.
- Consult infectious disease/public health.
- Report possible treatment failure.

Bottom Line

The WHO 2005 shigellosis guideline remains the core global WHO shigellosis-specific guideline, but its fluoroquinolone-first approach must now be modified by AMR surveillance and AST. As of **29 April 2026**, the most clinically defensible WHO-based approach is: **rapid syndromic recognition of bloody diarrhoea, hydration-first care, selective antibiotic use for dysentery/severe/high-risk cases, AST-guided antimicrobial choice, avoidance of antimotility agents, strong WASH/outbreak-control measures, and active AMR surveillance.**